Fusion Data Card

Authors	Group Name	Created
tfs523, jeb6694	Fusion	8/31/26

Contents

1. Card ID
2. Dataset at a glance
3. Usage information
4. Source and collection process
5. Cleaning and processing
6. Exploration study
7. Evaluations

Card ID

Primary Dataset	*“On_Time_Reporting_Carrier_On_Time_Performance_(1987_present)_2026_6.csv”*
Description	*This dataset contains 607,577 US domestic flight records reported by 12 airlines across 351 airports, with scheduled and actual times, delays by cause, cancellations and diversions, from 06/01/2026 to 06/30/2026.*
Supplemental Dataset 1	*“nldn-tiles-202606.csv”*
Description	*This dataset contains 475,343 daily lightning-strike counts on a 0.1° latitude/longitude grid covering the continental US, totalling 4,538,490 strikes, from 06/01/2026 to 06/30/2026.*
Supplemental Dataset 2	*“hail-tiles-202606.csv”*
Description	*This dataset contains 338,241 daily radar-detected hail counts on the same 0.1° grid, totalling 2,428,076 detections, from 06/01/2026 to 06/30/2026.*

Add more rows to the above table for each additional supplemental dataset.

Datasets at a glance

Primary dataset

The primary dataset contains 607,577 examples representing individual US domestic flight segments (One row per flight) in the period from 06/01/2026 to 06/30/2026. It covers 12 reporting carriers and 351 origin and destination airports. In this period 1.65% of flights were cancelled and 0.39% diverted, with a mean departure delay of 20.7 minutes.

Fields (if applicable)

Field name	Data Type	Description
FlightDate	date (string)	Flight date, YYYY-MM-DD. 2026-06-01 to 2026-06-30. e.g. 2026-06-16
Reporting_Airline	string	Carrier code. 12 values: AA, AS, B6, DL, F9, G4, MQ, OH, OO, UA, WN, YX
Origin / Dest	string	Origin/destination airport code. 351 distinct. e.g. MKE, DTW
CRSDepTime	integer	Scheduled departure, local hhmm. 1–2359. e.g. 1649
DepTime	integer	Actual departure, local hhmm. 1–2400. Null if cancelled
DepDelayMinutes	float	Departure delay, minutes; negative values floored to 0. 0–2759
ArrDelayMinutes	float	Arrival delay, minutes. 0–2752
Cancelled	float (0/1)	1 if cancelled. 10,019 flights (1.65%)
CancellationCode	string	Reason: A=Carrier, B=Weather, C=National Air System, D=Security
Diverted	float (0/1)	1 if diverted. 2,363 flights (0.39%)
WeatherDelay	float	Minutes attributed to weather. 0–1823
NASDelay	float	Minutes attributed to the National Air System. 0–1310
CarrierDelay	float	Minutes attributed to the carrier. 0–2692
LateAircraftDelay	float	Minutes from a late inbound aircraft. 0–2398
Distance	float	Great-circle distance, miles. 31–4983

Supplemental dataset 1

The supplemental dataset contains 475,343 examples representing one grid-cell-day of lightning activity each, across 129,733 distinct 0.1-degree cells spanning longitude −125.6 to −44.5 and latitude 16.6 to 51.7, in the period from 06/01/2026 to 06/30/2026.

Fields (if applicable)

Field name	Data Type	Description
ZDAY	date (integer)	UTC day, YYYYMMDD. 20260601–20260630. e.g. 20260601
CENTERLON	float	Grid cell centre longitude, degrees. −125.6 to −44.5. e.g. -96.0
CENTERLAT	float	Grid cell centre latitude, degrees. 16.6 to 51.7. e.g. 37.6
TOTAL_COUNT	integer	Lightning strikes in that cell that day. 1–1,481, mean 9.5

Supplemental dataset 2 (if applicable)

The supplemental dataset contains 338,241 examples representing one grid-cell-day of radar-detected hail each, across 93,653 cells spanning longitude −167.8 to 147.9 and latitude 10.5 to 67.4, in the period from 06/01/2026 to 06/30/2026.

Fields (if applicable)

Field name	Data Type	Description
ZDAY	date (integer)	UTC day, YYYYMMDD. 20260601–20260630
CENTERLON	float	Grid cell centre longitude, degrees. −167.8 to 147.9. e.g. -79.9
CENTERLAT	float	Grid cell centre latitude, degrees. 10.5 to 67.4. e.g. 33.0
TOTAL_COUNT	integer	Hail detections in that cell that day. 1–228, mean 7.2

If more than two supplemental datasets will be used, provide the same details for each one. If only one supplemental dataset is being used, remove the supplemental dataset 2 section.

Additional notes

- hail-tiles-202606.csv misdescribes itself in its own header comment as "the count of Vaisala NLDN lightning strikes" — identical boilerplate to the lightning file. This is a copy-paste error in NOAA's header generation, present across their hail, structure and lightning tile products. The data are hail detections; only the column names and filename are reliable.

- The five delay-cause fields are populated for only 153,037 of 607,577 rows, because BTS attributes causes only when arrival delay reaches 15 minutes. These nulls are not missing at random, and treating them as zero is a modelling decision rather than a cleaning step.

- The primary CSV's header row ends in a trailing comma, producing an empty 110th column that pandas names Unnamed: 110.

- Spatial extents differ between the supplemental sets: lightning covers the continental US, while hail extends to Alaska, Pacific and Caribbean radar sites and crosses the antimeridian. National maps need consistent extents.

Usage information

Questions driving the exploration study

1. How does the daily volume of lightning near an airport relate to that airport's mean departure delay?
2. Which of the 351 airports show the strongest weather-delay sensitivity, and does that track their geography?
3. Does BTS's own WeatherDelay attribution agree with independently observed lightning and hail activity on the same date and location?
4. How do the five delay causes shift in composition on days with severe weather versus clear days?
5. Do hail days and lightning days produce different operational responses (Cancellations versus delays)?
6. How far does weather disruption propagate through LateAircraftDelay to airports with no local severe weather?
7. Which carriers absorb weather disruption best, controlling for their route geography?

Sanctioned Use(s)

- BTS On-Time Performance: collected under 14 CFR Part 234 for public reporting of airline service quality, and published to support the DOT's Air Travel Consumer Report. Public analysis and research are explicitly sanctioned uses.

- NOAA SWDI (both supplemental sets): NCEI publishes the Severe Weather Data Inventory as a research and analysis archive of severe-weather records.

- Discouraged: SWDI hail, mesocyclone and tornado-signature products are radar-algorithm output, not verified ground reports — the raw point files label themselves #This file contains experimental data. They should not be treated as confirmed hailfall, nor substituted for NCEI's Storm Events Database where ground truth matters. Coverage is limited to NEXRAD radar range.

Licensing

- BTS: a work of the US federal government, in the public domain under 17 U.S.C. §105. No restrictions on sharing, transformation or reuse. BTS requests citation as the source.

- NOAA SWDI: NOAA open data — free to use and redistribute, with attribution requested for unaltered data.

- The lightning data originates from Vaisala's commercially-operated NLDN. Under NOAA's contract, raw flash and strike-level data are restricted to government and military users. The 0.1-degree daily tile aggregate we use is the publicly releasable derivative — that aggregation is why the tiles product exists. Our file is unrestricted, but strike-level detail cannot be reconstructed from it.

Privacy considerations

No personally identifiable information in any of the three datasets. The BTS data is operational and flight-level: no passenger, crew or booking records. Tail_Number identifies a specific airframe and no anonymisation or redaction was applied or required. The two SWDI datasets are geophysical observations aggregated to a spatial grid, with no human subjects.

Format

- The primary dataset is in a CSV file.
- The first supplemental dataset is in a CSV file.
- The second supplemental dataset is in a CSV file.

Source and collection process

Origin

- Primary: Reporting Carrier On-Time Performance (1987–present) table, "On-Time" database, BTS TranStats data library: https://www.transtats.bts.gov/ontime/

- Supplemental 1 and 2: NOAA NCEI Severe Weather Data Inventory, CSV archive: https://www.ncei.noaa.gov/pub/data/swdi/database-csv/v2/ (files nldn-tiles-202606.csv.gz and hail-tiles-202606.csv.gz)

Original datasets author/publisher

- Primary: US Department of Transportation, Bureau of Transportation Statistics, Office of Airline Information. Collected under 14 CFR Part 234 from carriers accounting for at least 0.5% of domestic scheduled passenger revenues — which is why only 12 airlines appear.

- Supplemental: NOAA National Centers for Environmental Information (NCEI), Asheville, NC. Lightning observations are sourced from Vaisala Inc.'s National Lightning Detection Network under contract; hail detections derive from the NEXRAD Level-III Hail Detection Algorithm.

Original datasets point of contact

- Primary: BTS TranStats support — https://www.transtats.bts.gov/
- Supplemental: NCEI Customer Engagement Branch (ncei.orders@noaa.gov)

Original metadata

- Primary: readme.html, shipped with the BTS extract; converted and stored at Datasets/Primary/README.md. Field definitions also at https://www.transtats.bts.gov/Fields.asp?gnoyr_VQ=FGJ
- Supplemental: SWDI documentation at https://www.ncei.noaa.gov/pub/data/swdi/database-csv/v2/ and https://www.ncei.noaa.gov/products/severe-weather-data-inventory

Collection process

- Primary: self-reported by airlines to DOT monthly under the Airline Service Quality Performance programme; released ~2 months after month end. June 2026 was published 12 August 2026. Details: https://www.bts.gov/topics/airlines-and-airports/understanding-reporting-causes-flight-delays-and-cancellations
- Supplemental: NLDN sensors detect cloud-to-ground flashes continuously; NEXRAD radars run the hail detection algorithm per volume scan. NCEI aggregates both to daily counts per 0.1-degree grid cell on UTC day boundaries.

Collection or access date

Primary dataset downloaded 31 August 2026. Supplemental datasets downloaded 31 August 2026.

Refresh date/schedule

- Primary: monthly, roughly two months in arrears. June 2026 released 12 August 2026; July expected mid-September. The file we hold is a fixed monthly extract and will not change.
- Supplemental: monthly files are posted early in the following month and revised — nldn-tiles-202606 was written 5 July 2026, and NCEI's directory shows August 2026 files updated as recently as 31 August 2026. Our copies are point-in-time snapshots taken 31 August 2026.

Cleaning and processing

Missing values

Describe the methods used to identify null or empty entries and the specific strategy applied to handle them.

Aggregation

Detail how data was summarized or grouped (e.g., by time periods, geographic regions, or categories) to achieve the necessary level of granularity for your analysis.

Joins of input sources

Explain the logic used to combine the primary and supplemental datasets, including the common keys (e.g., unique IDs or timestamps) and the type of join performed.

Exploration study

Descriptive statistics

Add basic statistics for each field here, as relevant, e.g. some statistics will be relevant for numeric data, but not for strings.

Distribution plots

Include charts showing each key field’s distribution.

Visual analysis

Select and include key charts that best represent the primary insights uncovered during your data exploration.

Streamlit application

Select and include key pages that let users explore your core questions interactively.

Vis Agent samples

Include a few representative visualizations from your vis agent, each with its prompt. Choose examples that show the agent's range; save any assessment for the Agent evaluation study.

Evaluations

First user study

Summarize what users reported learning about the data through your visual analysis. Group recurring themes rather than listing every comment.

List the questions your users still had after exploring the application.

Record the suggestions users offered for making the visualizations clearer or more useful.

Agent evaluation study

Summarize the test questions and your observations of the agent's behavior across runs. Document three areas of improvement grounded in those observations.

Second user study (extra credit)

List the questions users asked your Vis Agent.

Summarize what they learned about the data. Group recurring themes rather than listing every comment.

Record the suggestions users offered for making the agent more functional and useful.